

Hospital Variation in Episiotomy Use and the Risk of Perineal Trauma During Childbirth

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ABSTRACT: Background: *Episiotomies are the most frequently performed surgical procedure among United States women, but there are no published epidemiological studies of the extent of variation in episiotomy use among different obstetrical providers, or the outcomes associated with different levels of use. The objective of this study was to assess the extent of hospital variation in the use of episiotomy and the relationship between hospital episiotomy use and the incidence and risk of perineal trauma among women residing in a large urban area in the United States.* **Methods:** *Linked birth certificate and hospital discharge data pertaining to births to nulliparous women without prolonged labor or obstructed deliveries, and with infants weighing between 2500 and 4000 g, were analyzed for 18 major maternity hospitals. The relationship between episiotomy use and perineal trauma at the hospital level was examined using regression analysis.* **Results:** *Hospital episiotomy rates ranged from approximately 20 to 73 percent. The rate of third or fourth degree perineal lacerations varied from a low of 4 percent to a high of more than 13 percent among hospitals. Rates of episiotomy were significantly correlated with rates of a third or fourth degree perineal laceration ($r = 0.70$; $p < 0.01$), and with the hospital-specific, adjusted odds ratios for such lacerations ($r = 0.65$; $p < 0.01$).* **Conclusions:** *Findings from the analysis of epidemiological data for this study population were consistent with those from clinical studies, indicating that liberal as opposed to restrictive use of episiotomy is unwarranted, and probably even harmful. Hospital episiotomy rates exceeded 20 percent in all cases, but such rates appear difficult to justify in face of the evidence. (BIRTH 29:2 June 2002)*

Episiotomies occur in an estimated 35.2 percent of all vaginal deliveries in the United States, and represent the most frequent type of surgery performed on the female population (1). Research indicates that physicians who practice prophylactic or liberal as opposed to more restrictive use of the procedure obtain none of the purported benefits in terms of pelvic floor protection or avoidance of damage to the perineum during childbirth; moreover, in general, episiotomy patients incur greater risks with respect to perineal trauma and excessive maternal blood loss

from delivery, and may experience more pelvic pain, sexual dysfunction, and urinary incontinence in the postpartum period (2–14). Although episiotomy use has declined somewhat in the last two decades, many physicians continue to perform the procedure routinely (15). This would appear difficult to justify in view of the evidence; some observers have, in fact, gone so far as to characterize the continued, widespread use of the procedure as a form of violence against women (16,17).

It has been suggested that episiotomy rates of more than 20 or 30 percent are excessive (5,18), but surprisingly little is known about when, where, why, by whom, and with what consequences such rates are typically exceeded (19). Epidemiological investigations of the extent of variation in episiotomy use among different hospitals or physicians, or of the outcomes associated with liberal as opposed to restrictive use by obstetrical providers, are conspicuously absent from the published research literature.

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The current study focused on episiotomy use for a large population of childbearing women residing in a single, major urban center in the United States. Levels of episiotomy use and patterns of maternal morbidity were examined for deliveries occurring at any one of 18 local hospitals serving the Philadelphia metropolitan area. The objective was to assess the extent of variation from hospital to hospital in the frequency of episiotomy use, and the extent to which any significant hospital-level variation in use was correlated with the incidence and estimated risk of perineal trauma among women delivering at different hospitals.

Methods

Study Sample

The source of data for this study was the Philadelphia Perinatal Database, which was created and is maintained by the Philadelphia Department of Public Health. As described elsewhere (20), it contains computer merged data elements extracted from both electronic birth and hospital discharge summaries for all Philadelphia resident deliveries occurring between June 1, 1994, and July 31, 1998. Computerized matching algorithms resulted in the linking of maternal birth and hospital discharge records for more than 87 percent of all Philadelphia resident live births for this time period. An analysis of the birth record data pertaining to deliveries for whom hospital discharge data did not exist or could not be linked (because of missing or out-of-range values in fields used for matching) indicated that "non-matched" records were similar to "matches" in terms of sociodemographic characteristics, pregnancy risk factors, birth outcomes, labor and delivery complications, and modes of delivery.

Based on procedure and diagnostic codes appearing and reported by hospitals on the hospital discharge record, we calculated, compared, and correlated rates of episiotomy use with the rates and with estimates of the relative risk of third or fourth degree lacerations for each of the study hospitals. Third and fourth degree lacerations represent a serious form of maternal morbidity associated vaginal delivery, and their frequency of occurrence tends to increase when obstetrical interventions such as vacuum extraction, forceps, or both are used. Women who incur such lacerations during childbirth are at much higher risk for subsequent development of anal sphincter disruption and damage than women who do not incur such injuries (21).

We restricted the study to the 18 largest inner city hospitals where at least 300 resident births occurred

between June 1, 1994, and July 31, 1998. More than 86 percent of all resident live birth deliveries occurred at one of these 18 hospitals during this period.

We limited the analyses to nulliparous women because they are known to be at far greater risk of third or fourth degree lacerations during childbirth than multiparas. Preliminary analyses of the data indicated that, among the nulliparas in the database, instrumental delivery, high infant birthweight, prolonged labor, and the presence of obstructed delivery due to malpresentation or fetopelvic disproportion were all significant medical risk factors for third or fourth degree lacerations. The prevalence of these factors among the study participants varied somewhat from hospital to hospital. To eliminate or minimize the influence of these factors on any observed relationship between hospital episiotomy rates and our measure of perineal trauma, all subsequent analyses were restricted to only those deliveries involving spontaneous vaginal births and infants weighing between 2500 and 4000 g. For similar reasons deliveries with ICD-9 codes 660.0 through 660.4 appearing on the hospital discharge record, indicating obstructed labor due to malposition or disproportion, were excluded from the study sample. If precipitous or prolonged labor was indicated on either the birth record (labor < 3 hr or > 20 hr) or the hospital discharge record (ICD-9 codes 661.3, 662.0, 662.1, or 662.2), the delivery was also excluded from the study sample. In total, using these criteria, 1.6 percent of all normal birthweight, spontaneous vaginal deliveries were identified as "obstructed," and 1.5 percent were classified as either "prolonged" or "precipitous," and hence were excluded.

Of the total 49,692 spontaneous vaginal births occurring during the above-mentioned period 16,722 (34%) were to nulliparas. Of those, 14,292 (85%) remained eligible and were analyzed for the purposes of the current study.

Data Analysis

To quantify the relationship between episiotomy use and extent of perineal trauma, we calculated, for each hospital, rates and adjusted odds ratios of third or fourth degree lacerations. Odds ratios were adjusted for "case-mix" factors, including maternal age, race/ethnicity, education, and infant birthweight. The distribution of case-mix variables across hospitals and the relationship of these variables to laceration rates are not presented, but are available from the authors on request. Ordinary least squares regression was used to determine the bivariate correlations between (a) hospital episiotomy rates and laceration rates, and (b) hospital episiotomy rates and the

adjusted odds ratios (risk of severe laceration) for each hospital. Logistic regression analysis was used to calculate all adjusted odds ratios. All calculations and analyses were conducted on a desktop personal computer using software available in the Statistical Package for Social Sciences (22).

Results

Descriptive information about the study sample is presented in Table 1. Nearly 3 of 5 (57.9) deliveries were to African American women, slightly more than 2 of 5 (42.5%) were to adolescent mothers, and almost one-third (31.9%) were to women who had not completed high school at the time of birth. Almost 42 percent of all women underwent episiotomy, and 7.7 percent incurred a third or fourth degree laceration during childbirth. Overall the rate of severe tears was highest for women undergoing episiotomy: 9.9 percent of those with episiotomy had a third or fourth degree tear, compared with 6.0 percent of those without the procedure (not shown).

Episiotomy rates varied enormously, from a low of 20.3 percent in Hospital A to a high of 73.0 percent in Hospital R (Table 2). Three of the 18 hospitals performed episiotomies on most nulliparas in the study, even though none of the deliveries in the study sample involved macrosomic infants, or prolonged or

Table 1. Description of Study Sample

Characteristic	No.	(%)
All women	14,292	(100.0)
Race/ethnicity		
White	3,640	(25.5)
Black	8,262	(57.8)
Hispanic	1,595	(11.2)
Asian/other	795	(5.6)
Mother's age (yr)		
< 19	6,081	(42.5)
20–24	3,974	(27.8)
25–29	2,441	(17.1)
30–39	1,717	(12.0)
40+	79	(0.6)
Mother's education		
< High school	4,546	(31.8)
High school	6,244	(43.7)
Beyond high school	3,502	(24.5)
Infant birthweight (g)		
2500–2999	4,994	(34.9)
3000–3499	6,031	(42.2)
3500–3999	3,267	(22.9)
With episiotomy		
Yes	5,980	(41.8)
No	8,312	(58.2)
Third/fourth degree laceration		
Yes	1,098	(7.7)
No	13,194	(92.3)

Table 2. Episiotomy Rates and Risk of Third or Fourth Degree Perineal Lacerations for 18 Study Hospitals

Hospital	No. of Births	Episiotomy Rate (%)	Rate (%)	Laceration	
				Adj Odds Ratio*	95% CI
A	759	(20.3)	(4.0)	1.0	reference
B	280	(28.1)	(6.7)	2.99	0.98, 9.12
C	792	(31.8)	(6.8)	3.30	1.17, 9.34
D	164	(31.8)	(2.7)	1.73	0.59, 5.02
E	1356	(32.2)	(6.0)	2.94	1.05, 8.22
F	2107	(33.3)	(9.3)	3.61	1.32, 9.90
G	551	(33.9)	(4.5)	1.51	0.51, 4.48
H	364	(34.6)	(5.2)	2.21	0.73, 6.70
I	628	(41.0)	(6.5)	2.95	1.03, 8.45
J	1662	(44.0)	(9.3)	4.18	1.52, 11.50
K	537	(44.5)	(5.5)	2.68	0.91, 7.87
L	1138	(45.2)	(10.5)	4.47	1.62, 12.36
M	524	(45.6)	(5.7)	2.26	0.78, 6.60
N	845	(48.8)	(8.2)	4.38	1.55, 12.41
O	571	(54.7)	(6.5)	2.44	0.85, 7.06
P	270	(59.4)	(10.2)	3.59	1.22, 10.59
Q	1095	(63.2)	(8.3)	3.82	1.37, 10.63
R	649	(73.0)	(13.3)	5.07	1.81, 14.18

* Adjusted relative odds of third or fourth degree perineal tear, based on logistic regression analysis controlling for mother's age, race/ethnicity, and education, as well as infant birthweight.

obstructed labor. In all but two of the study hospitals, episiotomy rates exceeded 30 percent.

The incidences of third and fourth degree lacerations for the study hospitals also ranged widely, from a low of 4 percent in one (Hospital A) to a high of 13.3 percent in another (Hospital R). Calculation and comparison of adjusted odds ratios for each hospital from the logistic regression analyses indicated that this variation was independent of differences related to patient sociodemographic and infant birthweight characteristics. Adjusted for such factors, the results indicated that the relative risk of incurring a third or fourth degree laceration was up to 5 times greater, depending on the hospital where the delivery occurred.

Both the calculated rates and estimated risks of third or fourth degree lacerations were significantly related to the frequency of episiotomy use (Table 3). The bivariate regression correlation coefficient for the relationship between rates of episiotomy and simple rates of third and fourth degree lacerations was 0.70 ($F_{1,16} = 15.5$; $p < 0.01$). For the relationship between rates of episiotomy and the adjusted odds ratios, the coefficient was 0.65 ($F_{1,16} = 11.5$; $p < 0.01$). A simple scatter plot of these values (Fig. 1) revealed a clearly discernible linear pattern, with a positive association between the level of hospital episiotomy use and the estimated relative risk of perineal trauma among women in each of the respective hospitals.

Discussion and Conclusions

We found substantial differences from hospital to hospital in terms of the frequency with which episiotomies were performed on low-risk nulliparous women residing within a single, relatively large urban area in the United States. Since parity is arguably the single most important risk factor for episiotomy use, and the study sample was restricted to women with no previous deliveries, it seems reasonable to hypothesize that the wide range in rates observed is largely a function of the policies, procedures, preferences, or “practice styles” of individual physicians and hospitals. Subsequent analyses of additional information available in the database lent some support for this view. More specifically, the hospitals in this study with the highest episiotomy rates tended also to be those with the highest rates of vacuum/forceps use, cesarean deliveries, labor augmentation and induction for nulliparas. This suggests different “institutional cultures” with respect to the use of obstetrical interventions, with “high-intervention” hospitals leaning toward more liberal use of episiotomy to assist vaginal delivery.

The most important study finding was that more liberal episiotomy use was highly correlated with the hospital-specific rates and estimated risks of perineal trauma. This result is consistent with findings from two other studies indicating that serial reductions in the frequency of episiotomy use are associated with a decrease in the incidence of perineal trauma (3,23). In the context of these as well as other related findings, the results in the current study build on a growing body of evidence revealing the lack of any tangible benefits, and indeed the potential harm, associated with the liberal or routine use of a commonly performed obstetrical procedure.

Although no established guidelines or official targets pertaining to any ideal episiotomy rate for

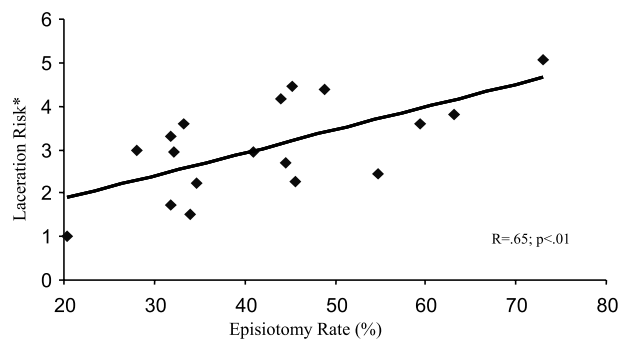


Fig. 1. Relationship between episiotomy rate and risk of third or fourth degree perineal laceration for 18 study hospitals. *Adjusted relative odds of third or fourth degree perineal tear, based on logistic regression analysis controlling for mother's age, race/ethnicity, and education, and infant birthweight.

any given population are available, it is of note that, in this study, the hospital with the lowest episiotomy rate of about 20 percent had the lowest rate of third and fourth degree lacerations, and women appeared to have the lowest estimated relative risk for incurring such lacerations.

No other similar epidemiological studies, to our knowledge, pertaining to any other large population of childbearing women residing elsewhere in the United States, have compared relevant outcomes for hospitals with widely varying episiotomy rates. Additional studies are warranted to determine if these findings can be generalized to other populations.

Our findings highlight the need for clarification of the appropriate medical indications for episiotomy use, and indicate the need for more systematic, ongoing assessments of the outcomes associated with different levels of episiotomy use by different obstetrical practitioners. In many hospital settings episiotomies have been performed so frequently for so long that the “benefits” are simply taken for granted; despite good intentions, many physicians find it difficult to abandon the procedure that they have been taught is routine, and that is widely accepted among colleagues as an effective way to assist even an uncomplicated birthing process “safely” (24–28). The evidence from clinical trials contradicting such assumptions is unlikely, by itself, to change physician practices (24). The relative disadvantages of frequent episiotomy use may have to be documented, communicated, and addressed multiple times in multiple ways before systematic and sustained reductions in use will occur.

Although the procedure remains deeply rooted in the practice of obstetrics in the United States, the trend toward increasing episiotomy rates has been reversed (15). This reversal probably reflects a

Table 3. Results of Bivariate Regression Analysis of Relationship Between Hospital Episiotomy Rates and Rates and Risk of Severe Perineal Lacerations ($n = 18$)

Measure	Episiotomy Rate and Rate of Third or Fourth Degree Laceration	Episiotomy Rate and Relative Risk of Third or Fourth Degree Laceration
Correlation coefficient (r)	0.70	0.65
Percent of variation explained (r^2)	0.49	0.42
p value	0.0012 ($F_{1,16} = 15.5$)	0.0037 ($F_{1,16} = 11.5$)

generalized erosion of a previously unshakeable belief in the benefits of routine episiotomy use among physicians—which is at least to some extent attributable to the cumulative weight of evidence from various studies conducted both within and outside the United States. Although randomized controlled studies are considered to be the “gold standard” in the field, population-based, observational studies such as the current one also have their place. As is the case here, they can confirm the generalizability and relevance of findings from randomized controlled trials. Moreover, “local data,” available at the county, hospital, or even physician level, can provide the baseline information necessary to document and compare obstetrical practices. The results of the analyses of such data can act as a catalyst for critical evaluation of the outcomes associated with specific obstetrical interventions and with varying levels of use by different providers (28). Routine episiotomy use persists despite the scientific evidence contrary to its efficacy; yet practitioner behavior with respect to its use should probably be regarded as difficult rather than impossible to change. Attempts to reduce episiotomy use by raising awareness of procedure rates and related outcomes among practitioners (29), or by setting goals for reduction in use as part of an overall quality improvement program (23), for example, have met with some success.

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